SAVANA

Reliable Analysis of Somatic Structural Variants and Copy Number Aberrations Using Long-Read Sequencing

Elrick, Sauer*, Espejo Valle-Inclan et al.*

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Computational Biology — Scribe Presentation
Mohammad Awawda



Outline

1. **Background**
   The problem of somatic SV detection in cancer genomes
2. **SAVANA Algorithm**
   Machine learning approach for SV/SCNA detection in long reads
3. **Benchmarking Results**
   Replicate analysis and cross-platform comparison
4. **Key Figures**
   Replicate concordance, specificity, and clinical validation
5. **Conclusions**
   Impact and critical perspective



Background: Structural Variants in Cancer

Cancer genomes harbor diverse structural variants (SVs)

Deletions, inversions, duplications, translocations, and complex rearrangements shape cancer evolution

SVs drive cancer by inactivating tumor suppressors and amplifying oncogenes

Most cancer driver mutations are underpinned by SVs and somatic copy number aberrations (SCNAs)

Short-read sequencing has limited power to detect SVs

Breakpoints in repetitive regions cannot be unambiguously aligned — the SV landscape remains partially mapped

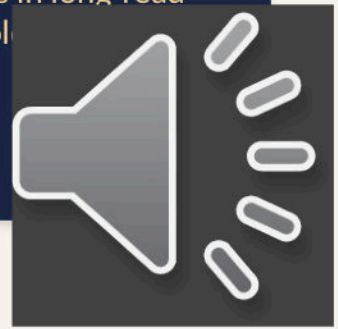
The Gap

Long reads can span repetitive regions

But existing SV callers yield high false-positive rates

No large-scale benchmarking on clinical samples existed

Unclear if high SV rates in long-read studies reflect true biol



Authors & Study Design

Research Group

Cortes-Ciriano Lab
EMBL-EBI, Hinxton, Cambridge, UK

Led by Adrienne Flanagan (UCL) & Isidro Cortes-Ciriano (EMBL-EBI)
Co-first: Hillary Elrick & Carolin M. Sauer

Dataset

99 tumor-normal pairs
57 soft-tissue sarcomas, 28 osteosarcomas, 14 glioblastomas

Matched Illumina + nanopore WGS
ONT: 51× tumor / 34× normal; Illumina: 118× / 41×

What Makes This Paper Unique

ML-based SV filtering

Random Forest trained on cross-platform data to distinguish true SVs from artifacts

Novel benchmarking

Simulated replicates + read-backed phasing for unbiased genome-wide assessment

Integrated

Combines SV, SCNA, estimation — works matched normal



SAVANA: Algorithm Overview

1 →

Breakpoint Discovery

Scan tumor & normal reads for SV-supporting split/gapped alignments; cluster nearby breakpoints

2 →

ML Classification

Random Forest on 70 features (MAPQ, read count, SV length, depth, etc.) — trained via cross-platform Illumina/ONT labels

3 →

Copy Number Analysis

CBS segmentation + SV breakpoints → genome-wide SCNA profiling

4

Purity & Ploidy

BAF at LOH regions → purity estimate; grid search over purity-ploidy space → allele-specific CN



Connection to Computational Biology Course

Sequence Alignment

SAVANA relies on read alignment (minimap2) and parses CIGAR strings for split/gapped alignments — core concepts from sequence alignment algorithms

Machine Learning on Sequences

Random Forest classification of breakpoints uses 70 features extracted from alignment data — supervised learning on genomic features

Segmentation Algorithms

Circular Binary Segmentation (CBS) for copy number detection — a changepoint detection algorithm on sequential data

Statistical Testing

Read-backed phasing uses binomial tests; relies on Wilcoxon tests and replicate ana statistical methods in bioinformatics

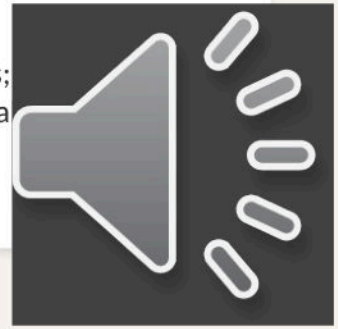


Figure 2: Replicate Benchmarking Strategy

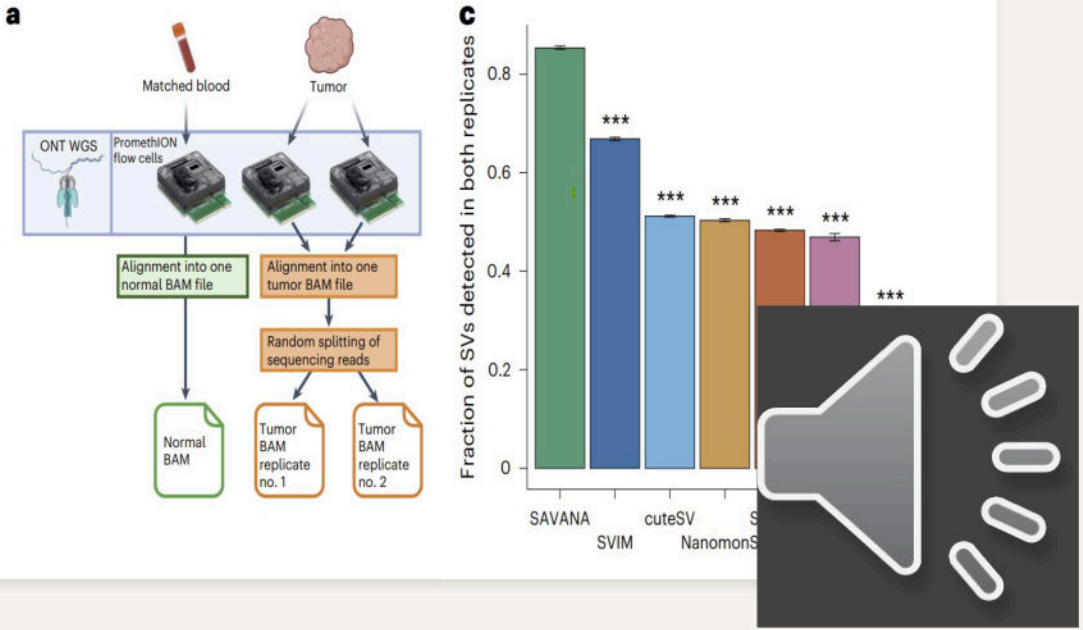
Method: Simulated Replicates (Fig. 2a–c)

Approach:

Split tumor reads randomly into 2 BAMs → run each caller independently → compare SV calls across replicates

Logic:

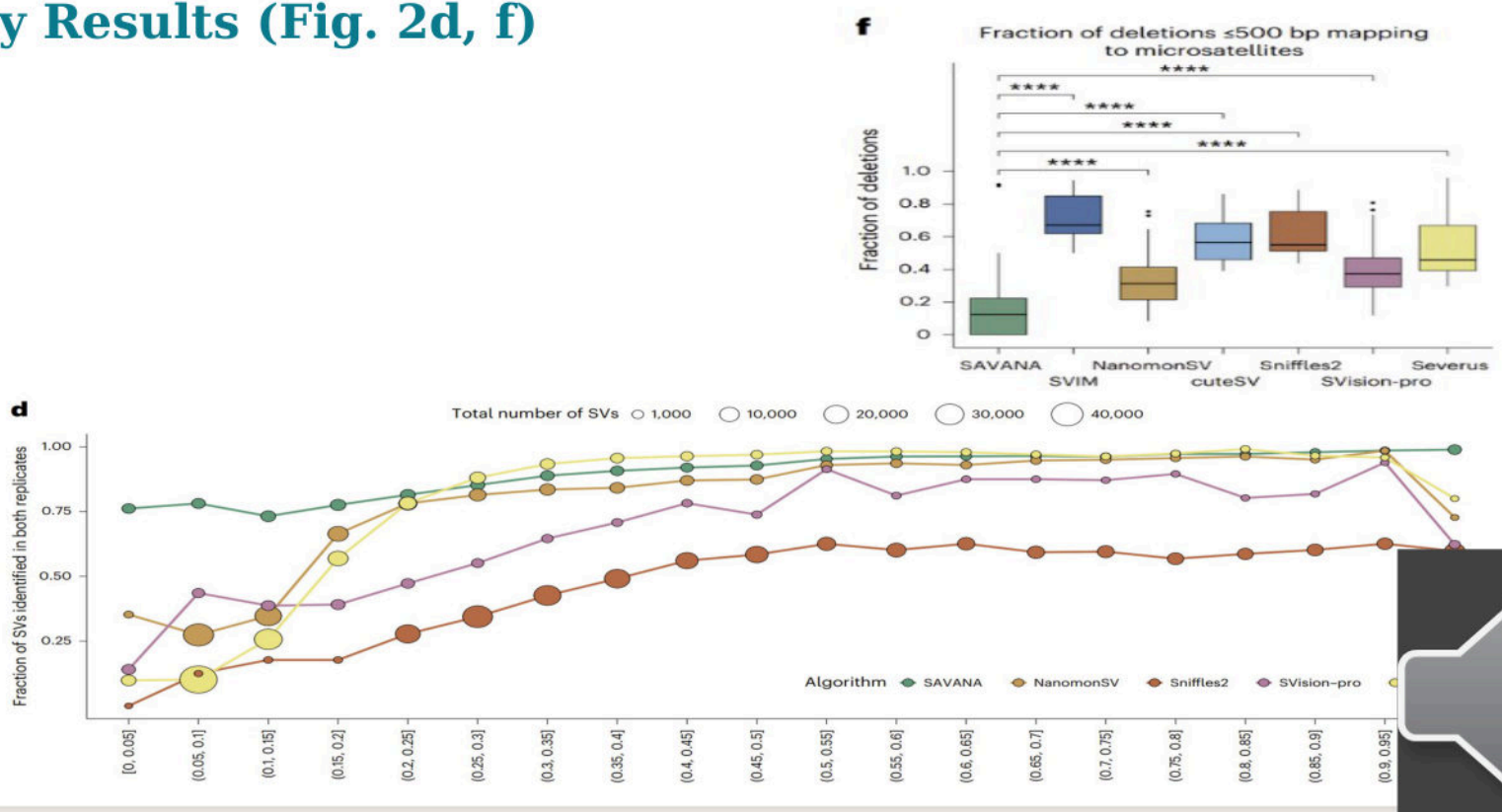
True SVs → detected in both replicates
False positives → detected in only one replicate



Source: Elrick & Sauer et al., Nature Methods 2025, Fig. 2

Figure 2: Replicate Benchmarking Strategy cont.

Key Results (Fig. 2d, f)



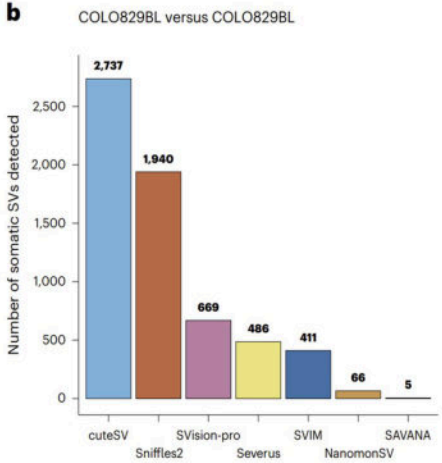
Source: Elrick & Sauer et al., Nature Methods 2025, Fig. 2

Figure 3: Quantifying Specificity

Normal vs. Normal Test (Fig. 3a–b)

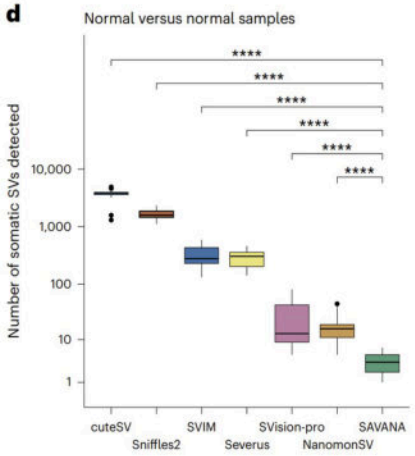
Experiment design:

Run algorithms using one normal replicate as "tumor" and another as "normal" — an ideal caller should report zero somatic SVs



Panel 3b — COLO829BL results:

X-axis: SV detection algorithms
Y-axis: number of false-positive somatic SVs



Panel 3d — 37 clinical blood samples:

Box plot showing false-positive SVs per sample
SAVANA: consistently lowest, often <5 FPs
Other tools: tens to thousands of false positives

False Positive Rates (COLO829BL)

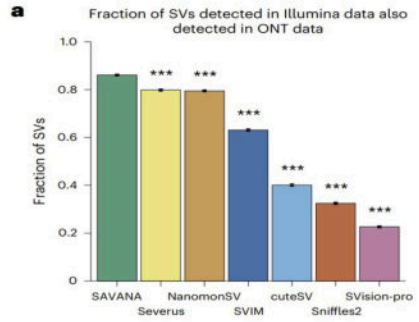
SAVANA	5
NanomonSV	66
SVIM	411
Severus	486
SVision-pro	669
Sniffles2	
cuteSV	

13× to 547×
FP rates than

Source: Elrick & Sauer et al., Nature Methods 2025, Fig. 3

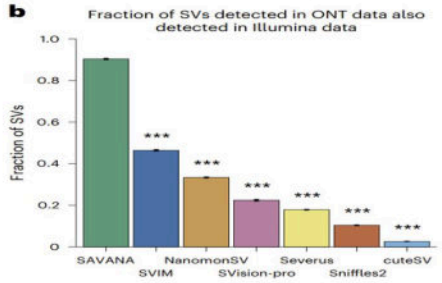
Figure 5: Long-Read vs. Short-Read Comparison

Panel 5a



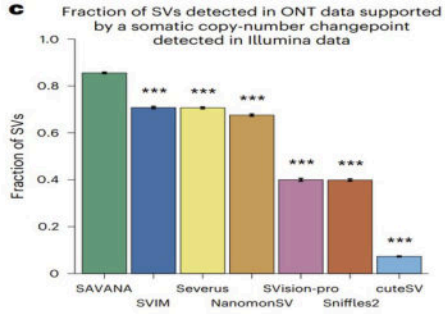
SAVANA: ~86% recall of short-read SVs

Panel 5b



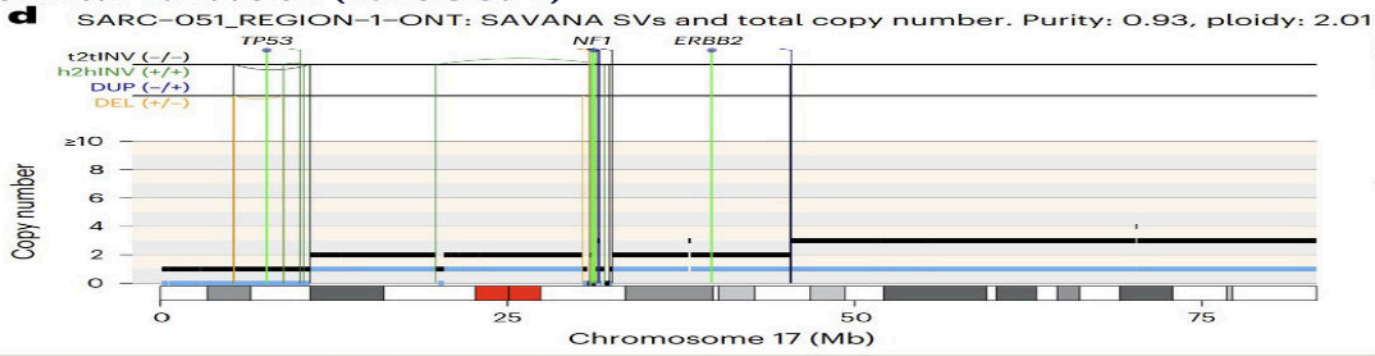
Other tools: <50% validated by Illumina

Panel 5c



SAVANA SVs match CN breakpoints

Clinical Validation (Panels 5d-f)



SAVANA detected 92% of Illumina-identified driver SVs in cancer genes (71/77)



Source: Elrick & Sauer et al., Nature Methods 2025, Fig. 5

Conclusions & Critical Perspective

Main Takeaways

- SAVANA enables reliable somatic SV & SCNA detection from long-read data
- 13–547× higher specificity than competing methods
- Most Illumina-detected SVs and driver mutations recovered
- Tumor-only mode enables clinical use without matched normal
- Replicate benchmarking is a best practice for SV caller evaluation

Critical Perspective

Strengths:

Largest matched Illumina/ONT clinical dataset (99 pairs)

Rigorous benchmarking with multiple complementary strategies

Open-source, VCF-compliant, clinically relevant

Limitations:

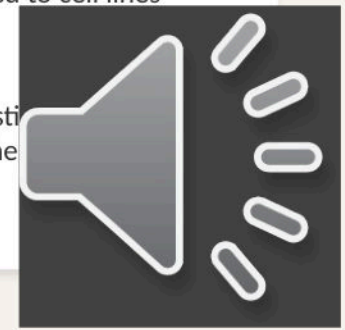
Training labels from Illumina — may miss long-read-only SVs

Limited to sarcomas & glioblastomas — generalization to other cancers not yet shown

ONT-focused; PacBio HiFi evaluation limited to cell lines

Broader Impact:

Suggests prior long-read SV studies overesti
due to artifacts — major implication for the



Thank You

Questions?

References

Elrick, Sauer "SAVANA: reliable analysis of somatic structural variants and copy number aberrations using long-read sequencing." Nature Methods 22, 1436–1446 (2025)

https://doi.org/10.1038/s41592-025-02708-0

Code: https://github.com/cortes-ciriano-lab/savana

All figures exported from Elrick & Sauer et al., Nature Methods 2025
The figures are taken from the paper pdf:
Figures 2,3,5 are taken from pages 5,6 and 8 respectively.

